

DHANUSH MAJJI

dmajji2003@gmail.com

+91 9491562335

[LinkedIn](#)

SUMMARY

Analyst skilled in Python, SQL, Excel, Tableau/Power BI, and statistical analysis. Experienced in data validation, reporting, and deriving insights for cross-functional teams. Familiar with ETL, basic modeling in Python/R, and data governance. Strong problem-solving mindset with a focus on process improvement in fast-paced fintech environments.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

Hyderabad, India

B.E. (Hons.) Chemical Engineering | Coursework (Minor) in Finance | GPA: 8.48/10.00 (First Division)

Nov 2020 - June 2024

- **Relevant Coursework:** Probability and Statistics, Fundamentals of Finance and Accounting, Derivatives and Risk Management, Security Analysis and Portfolio Management, Financial Risk Analytics and Management, Financial Management
- **Financial Risk Manager (FRM) Candidate** - Strong foundation in analytical methods, risk management, quantitative analysis

TECHNICAL SKILLS

Data Analytics & Tools	Python, SQL, R, Excel (pivot tables, power query), Power BI, Tableau
Modelling	TimeSeries Forecasting, Survival Analysis, Scenario Analysis, A/B Testing, Stress Testing
Data Science & ML	Linear & Logistic Regression, Lasso & Ridge, Decision Trees, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, Clustering, PCA, Neural Networks
Anomaly Detection	Isolation forest, One-Class SVM, Local Outlier Factor (LOF)
Libraries	pandas, numpy, scipy.stats, scikit-learn, statsmodels, cvxpy, yfinance, Keras, TensorFlow
Data Visualization	Matplotlib, Seaborn, Plotly, Excel Dashboards

EXPERIENCE

American Express lab for Data Analytics and Risk Technology, IIT Madras

Chennai, India

Machine Learning Project Assistant (Full-Time)

July 2024 - Dec 2024

- Developed a time-series aware, model-agnostic Explainable AI framework for fraud risk modelling, capturing sequential transaction patterns and enabling causally valid, regulator friendly counterfactuals with 94% fidelity.
- Designed a greedy search algorithm for optimal feature replacements, reducing false alarms by 18% and boosting fraud detection rates by 22%, enhancing overall risk management efficiency.
- Addressed limitations of traditional explainability methods (SHAP, LIME) in sequential data by building a framework that achieved 11% higher fidelity and 15% fewer undetected fraud cases (FNR), enabling more accurate fraud investigations.
- Improved fraud detection precision to 89% and recall to 86% by optimizing feature selection and classification thresholds

Aditya Birla Group

Mumbai, India

Data Scientist (Intern)

Jan 2024 - June 2024

- Developed minute-level prediction models from hourly data, enabling proactive quality control and real-time input tuning
- Built an adaptive AI system for real-time process optimization, reducing raw material costs by 15% and variability by 20%, delivering \$0.2-0.4 M in annual revenue gains, improving profit margins, cost efficiency, and smarter resource allocation.
- Collaborated with cross-functional teams including process engineers, data scientists, and plant managers across geographies (India, Spain, Canada) to deploy predictive analytics systems tailored to business needs.
- Engaged with internal stakeholders to gather requirements and convert them into analytical use cases, delivering insights that supported strategic decision-making.

RELEVANT PROJECTS

User Behavior Analytics | Independent Project

- Identified 3 onboarding friction points via funnel and feature adoption analysis using SQL & Python on clickstream data.
- Improved Day-7 retention by 12% through uplift modeling of personalized notifications and emails.
- Built cohort-based churn model using survival analysis and logistic regression to guide product improvements.

Credit Risk Modelling & Scorecard Development | Independent Project

- Built credit scorecards using logistic regression with WOE/IV binning, reject inference & optimal cut-off selection
- Modeled PD, LGD, EAD and aligned outputs with Basel norms, including capital estimation and macro stress testing
- Achieved 0.83 AUC, improved classification accuracy via threshold tuning, and generated actionable insights using Python & Excel.

Advanced Financial Analysis & Risk Forecasting | Course Project

- Estimated betas using CAPM across return frequencies for ABBOTINDIA; used ARIMA (2,0,2) to forecast daily returns.
- Forecasted conditional volatility using GARCH(1,1), enabling risk-sensitive decision-making

Business Insights Dashboard for Hospitality Management - Atliq Grands | Independent Project

- Developed a Power BI dashboard to track \$1.69 B revenue, occupancy rates, and \$298M in cancellations; identified Mumbai as the top-performing city (\$669M)
- Added interactive filters (city, room class, category, platform) to support data-driven decision-making

AWARDS & SCHOLARSHIPS/ MISCELLANEOUS

- MITACS Globalink Fellowship, Canada | 10,000 CAD | Selected among 500 out of 1.5M applicants
- PadmaSri B.V. Raju Best Merit Student Award | Awarded to the top performing student among 80 in Chemical Engineering dept.
- BITS Pilani MCN Scholarship | Awarded Semester II-VIII | One of 200 students
- Teaching Assistantship- Fundamentals of Finance & Accounting | Dept. of Economics & Finance, BITS Pilani | Cohort of 300 students